High Speed Railcar Suspension Dynamics Project Scope

Benjamin Sauer

High Speed Rail generally describes rail lines that support trains moving above 200 kilometers per hour (roughly 125 miles per hour). Rail lines are specially designed to be smoother and of higher quality than conventional rails. This is because at these velocities, high speed railcars experience more disturbances from the track and environment at a frequency particularly sensitive to humans. Ride comfort and stability are constrained by the railcar's ability to dampen these disturbances from the track and environment. This is the function of the suspension system.

Suspension systems for conventional passenger and freight rail utilize passive systems of springs and dampers to reduce vibrations from track and environmental disturbances. There are two suspension systems: the primary and secondary. The primary suspension connects the wheel/axle system to the rail bogie/trucks, which are the frames in which the wheels, traction, and braking systems are housed among other features. The secondary suspension is responsible for improving ride-comfort and railcar-body stability, and connects the rail bogie with the car-body.

High speed railcars utilize either semi-active suspension or active suspension to improve disturbance reduction beyond what a passive system is capable of. Semi-active suspension systems generally use special dampers with controllable viscosity/damping coefficients. Having a controllable component allows for adjustment of the damping coefficient to values optimal for minimizing vibrational response.

The present project will:

- Simulate the primary and secondary suspension dynamics of a high speed rail car using Python
- Apply control laws based upon a semi-active suspension system to minimize vibrational response from step and noise disturbances
- Model the suspension system, rail bogie, and simplified rail-car in Fusion 360 to confirm rail-car mass and dimensions assumptions.

This project is inspired by the Brightline West, a current high-speed rail project intended to connect Southern California to Las Vegas. Semi-active control was chosen as it is more affordable compared to active suspension. Affordability is crucial for the approval of any high speed rail project in the US, as high speed rail is new, experimental, and experiences difficulty in funding and implementation.

Currently, skyhook control is intended to be implemented on the secondary suspension vertical damping element. Skyhook excels at keeping vehicle bodies close to stationary thus improving ride comfort. Other control laws will be briefly examined. The control model will be assessed based on how well it maximizes ride comfort and railcar stability. Ride comfort will be assessed based on: RMS car-body acceleration, peak suspension travel, and frequency-weighted RMS acceleration per ISO 2631 (an international standard for evaluating human comfort in response to vibrational disturbances.

Assumptions will be made regarding rail car mass and dimensions prior to the CAD modeling phase so that the python simulation development can begin first. This data is taken from "Simplified Mechanical Description of AVE S-103 - ICE3 Velaro E High Speed Train" by José M. Goicolea. It should be noted that the data is unofficial and may not reflect the Velaro High Speed Train with complete accuracy. Nevertheless, the data should provide enough accuracy to guide development before confirmation via the CAD model. Below are some lists of the assumptions:

Main Body:

- Length of (intermediate) rail car: 24.775m
- Main body mass: 47800 kg
- Main body mass vertical moment of inertia: 1957888 kg*m^2

Bogie:

- Distance between bogie centers in one car: 17.375m
- Distance between axles in one bogie: 2.5m
- Bogie mass: 3500kg
- Bogie mass vertical moment of inertia: 4235kg*m^2
- Axle and wheelset mass: 1800kg

Primary Suspension:

- Lateral separation between suspension centers: 2m
- Vertical spring stiffness: 1200 kN/m
- Vertical viscous damping: 10 kN*s/m

Secondary Suspension:

- Lateral separation: 2m
- Vertical spring stiffness: 350 kN/m
- Vertical viscous damping: 20 kN*s/m

This model will only include vertical, roll, and pitch suspension and disturbance to simplify calculations during development. If time allows, lateral and yaw degrees of freedom will be added at the end. It should also be noted that wheel to rail contact dynamics will not be explicitly modeled, and the track will not be treated as a dynamic system, only as an input of disturbance. The structural flexibility of the railcar body will also be neglected.

See the references document for bibliography.